

ORF 524 — Final Exam

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Notes:

- Duration of examination: 180 minutes.
An additional 15 minutes are given to upload your solutions to Gradescope.
- All parts are equally weighted (5 points).
Please provide as much detail as possible in your answers.
Answers without proper justification will not receive full credit.
- Please start each question on a separate page.
A single PDF file must be submitted to Gradescope.
- Only one page of personal notes are allowed during the examination.
This is otherwise a closed-book and individual examination.

1 Best Linear Approximation (15 points)

Let Y be a scalar random variable and $\mathbf{X}$ be a random vector in $\mathbb{R}^d$. Assume that $\mathbb{E}[Y^2] < \infty$ and $g \in \mathcal{G} = \{g : \mathbb{E}[g(\mathbf{X})^2] < \infty\}$. We are interested in the predictor $g^* \in \mathcal{G}$ that minimizes the Mean Squared Error (MSE), defined as the risk functional $R(g) = \mathbb{E}[(Y - g(\mathbf{X}))^2]$. Let $\mu(\mathbf{X}) = \mathbb{E}[Y|\mathbf{X}]$.

1. *General Decomposition.* Show that $R(g) = \mathbb{E}[\text{Var}[Y|\mathbf{X}]] + \mathbb{E}[(\mu(\mathbf{X}) - g(\mathbf{X}))^2]$ for all $g \in \mathcal{G}$.
2. *Linear Model.* Assume that $\mu(\mathbf{X}) = \mathbf{X}^\top \beta_0$ for some unique parameter vector $\beta_0 \in \mathbb{R}^d$, and consider all candidates $g \in \mathcal{G}_L = \{g : g(\mathbf{X}) = \mathbf{X}^\top \beta, \beta \in \mathbb{R}^d\}$. Give sufficient conditions to ensure that $\mathcal{G}_L \subseteq \mathcal{G}$, and show that $R(g)$ is determined by a quadratic form involving the second moment matrix of the covariates, $\Sigma_{\mathbf{X}} = \mathbb{E}[\mathbf{X}\mathbf{X}^\top]$. Give sufficient conditions to ensure that μ is the unique solution of $\min_{g \in \mathcal{G}_L} R(g)$.

Since the first term in the general decomposition does not depend on β (irreducible error), minimizing the MSE is equivalent to minimizing this quadratic form. If $\Sigma_{\mathbf{X}}$ is positive definite, this quadratic form is strictly positive for all $\beta \neq \beta_0$, and zero if and only if $\beta = \beta_0$. Thus, β_0 is the unique minimizer under the (sufficient) condition that $\Sigma_{\mathbf{X}}$ is positive definite.

3. *Orthogonality Condition and Interpretation.* Consider now a general conditional mean $\mu(\mathbf{X})$, not necessarily equal to $\mathbf{X}^\top \beta_0$. What is the interpretation for the solution of $\min_{g \in \mathcal{G}_L} R(g)$?

2 Quasi-Maximum Likelihood and Poisson Regression (35 points)

Let $(Y_i, \mathbf{X}_i^\top)$ for $i = 1, \dots, n$ be independent and identically distributed observations, where $Y_i \in \mathbb{N}_0$ is a count response variable and $\mathbf{X}_i \in \mathbb{R}^p$ is a vector of covariates. Assume the conditional mean is correctly specified as:

$$\mathbb{E}[Y_i | \mathbf{X}_i] = \exp(\mathbf{X}_i^\top \boldsymbol{\beta}_0) \equiv \mu_i(\boldsymbol{\beta}_0),$$

where $\boldsymbol{\beta}_0$ is the population parameter vector. However, the population conditional distribution of $Y_i | \mathbf{X}_i$ is **not** necessarily Poisson. The conditional variance is denoted by $\text{Var}[Y_i | \mathbf{X}_i] = \sigma^2(\mathbf{X}_i)$.

This question considers two estimators for $\boldsymbol{\beta}_0$:

- *The Poisson Quasi-Maximum Likelihood Estimator:* $\hat{\boldsymbol{\beta}}_{\text{QMLE}}$, obtained by maximizing the Poisson log-likelihood despite potential misspecification of the distribution.
- *The Weighted Non-Linear Least Squares Estimator:* $\hat{\boldsymbol{\beta}}_{\text{WNLS}}$, obtained by minimizing a weighted sum of non-linear squared residuals.

Let $\nabla_{\mathbf{w}} g(\mathbf{w})$ and $\nabla_{\mathbf{w}}^2 g(\mathbf{w})$ denote the gradient and Hessian of the function $g : \mathbb{R}^d \mapsto \mathbb{R}$, respectively. Please answer the following questions.

1. *Poisson QMLE: Score.* Write down the Poisson log-likelihood function $\ell(\boldsymbol{\beta})$, that is, the log-likelihood under the assumption that $Y_i | \mathbf{X}_i \sim \text{Poisson}(\mu_i(\boldsymbol{\beta}_0))$. Derive the score function $\mathbf{S}_n(\boldsymbol{\beta}) = \nabla_{\boldsymbol{\beta}} \ell(\boldsymbol{\beta})$, and show that $\mathbf{S}_n(\boldsymbol{\beta}_0)$ has an expectation of zero, regardless of the population conditional distribution of $Y_i | \mathbf{X}_i$.
2. *Poisson QMLE: Hessian.* Derive the Hessian matrix $\mathbf{H}_n(\boldsymbol{\beta}) = \nabla_{\boldsymbol{\beta}}^2 \ell(\boldsymbol{\beta})$. Let $\boldsymbol{\Omega}_0 = -\mathbb{E}[\nabla_{\boldsymbol{\beta}}^2 \ell(\boldsymbol{\beta}_0)]$ and $\boldsymbol{\Sigma}_0 = \text{Var}[\mathbf{S}_n(\boldsymbol{\beta}_0)]$. Show that the information equality holds ($\boldsymbol{\Omega}_0 = \boldsymbol{\Sigma}_0$) under the assumption that $Y_i | \mathbf{X}_i \sim \text{Poisson}(\mu_i(\boldsymbol{\beta}_0))$. Explain briefly why $\boldsymbol{\Omega}_0 \neq \boldsymbol{\Sigma}_0$ in the general case.
3. *Poisson QMLE: Asymptotic Normality.* Using standard M-estimation theory (e.g., Taylor expansion of the score function), state the asymptotic distribution of $\sqrt{n}(\hat{\boldsymbol{\beta}}_{\text{QMLE}} - \boldsymbol{\beta}_0)$. Explicitly provide the “sandwich” form of the asymptotic covariance matrix $\mathbf{V}_{\text{QMLE}}$.
4. *Poisson QMLE: Robust Standard Errors.* Given your result in the previous part, propose a consistent estimator for the asymptotic covariance matrix $\mathbf{V}_{\text{QMLE}}$ that is valid even if the Poisson variance assumption fails. This is often called the Huber-Eicker-White or “sandwich” estimator.
5. *WNLS Estimator: Objective Function and FOC.* Consider the objective function:

$$Q_n(\boldsymbol{\beta}) = \sum_{i=1}^n w(\mathbf{X}_i) (Y_i - \exp(\mathbf{X}_i^\top \boldsymbol{\beta}))^2$$

where $w(\mathbf{X}_i) > 0$ are weights. Derive the first-order condition for $\hat{\boldsymbol{\beta}}_{\text{WNLS}}$.

6. *WNLS Estimator: Optimal Weights.* Determine the specific weight function $w^*(\mathbf{X}_i)$ that makes the WNLS estimator asymptotically equivalent to the Poisson MLE.
7. *Misspecification.* Suppose now that the mean function is actually misspecified, i.e., $\mathbb{E}[Y_i|\mathbf{X}_i] \neq \exp(\mathbf{X}_i^\top \boldsymbol{\beta})$ for any $\boldsymbol{\beta}$. Discuss the convergence limit of the Poisson QMLE $\hat{\boldsymbol{\beta}}_{\text{QMLE}}$. Does it converge to a meaningful parameter?

3 Uniform Consistency of the Kernel Density Estimator (50 points)

Let $X_1, \dots, X_n$ be independent and identically distributed (i.i.d.) random variables taking values in $\mathbb{R}$ with a common probability density function f . Consider the Kernel Density Estimator:

$$\hat{f}_n(x) = \frac{1}{nh_n} \sum_{i=1}^n K\left(\frac{x - X_i}{h_n}\right)$$

where $h_n > 0$ is the bandwidth sequence and $K : \mathbb{R} \rightarrow \mathbb{R}$ is the kernel function. We assume throughout that $h_n \rightarrow 0$ as $n \rightarrow \infty$.

We impose the following assumptions.

- The density f is bounded (i.e., $\sup_x f(x) \leq \|f\|_\infty < \infty$) and uniformly continuous on $\mathbb{R}$.
- The kernel K is supported on $[-1, 1]$, integrates to one, is symmetric about 0, is bounded (i.e., $\sup_u |K(u)| \leq \|K\|_\infty < \infty$), and is Lipschitz continuous with constant L_K (i.e., $|K(u) - K(v)| \leq L_K |u - v|$).

This question proves the uniform consistency of the kernel density estimator estimator over a compact interval D :

$$\sup_{x \in D} |\hat{f}_n(x) - f(x)| = o_{\mathbb{P}}(1), \quad \text{as } n \rightarrow \infty,$$

where $D = [0, 1]$ (without loss of generality), and under specific rates on h_n (to be set below). The following well-known exponential probability inequality will play a crucial role.

- **Theorem (Bernstein's Inequality).** Let $W_1, \dots, W_n$ be independent zero-mean random variables such that $|W_i| \leq M$ almost surely. Then,

$$\mathbb{P}\left(\left|\sum_{i=1}^n W_i\right| > t\right) \leq 2 \exp\left(-\frac{t^2/2}{\sum_{i=1}^n \text{Var}[W_i] + Mt/3}\right)$$

Please answer the following questions.

1. *Bias-Variance Decomposition.* Decompose the error $|\hat{f}_n(x) - f(x)|$ into a bias term (deterministic) and a stochastic term (involving deviations from the mean). Explicitly define $\mathbb{E}[\hat{f}_n(x)]$.
2. *Pointwise Bias.* Show that for any fixed x , the bias converges to zero. That is, show $\lim_{n \rightarrow \infty} |\mathbb{E}[\hat{f}_n(x)] - f(x)| = 0$.
3. *Uniform Bias.* Show that the bias converges uniformly to zero. That is, $\lim_{n \rightarrow \infty} \sup_{x \in D} |\mathbb{E}[\hat{f}_n(x)] - f(x)| = 0$.
4. *Stochastic Term.* Let $Z_n(x) = \hat{f}_n(x) - \mathbb{E}[\hat{f}_n(x)]$, and express this statistic as a sum of independent zero-mean random variables $Y_i(x)$. Explicitly write out the form of $Y_i(x)$.

where $Y_i(x) = \frac{1}{nh_n} \left(K\left(\frac{x-X_i}{h_n}\right) - \mathbb{E}\left[K\left(\frac{x-X_1}{h_n}\right)\right] \right)$.

5. *Bounding the Variance and Magnitude.* To apply a concentration inequality, we need bounds on the variance and the absolute magnitude of the terms $Y_i(x)$.

(a) Show that $|Y_i(x)| \leq \frac{2\|K\|_\infty}{nh_n}$.

(b) Show that $\text{Var}[Y_i(x)] \leq \frac{\|f\|_\infty \|K\|_2^2}{n^2 h_n}$ (or a bound of order $O(\frac{1}{n^2 h_n})$).

6. *Exponential Concentration.* Apply Bernstein's inequality to $Z_n(x)$ to find an upper bound for $\mathbb{P}[|Z_n(x)| > \epsilon]$. Your bound should be of the form $2 \exp(-C \cdot nh_n \epsilon^2)$ for small ϵ .

7. *Discretization (Covering Argument).* Since the supremum is over an uncountable set $D = [0, 1]$, we cannot simply sum probabilities. We must discretize: construct a grid $x_1, \dots, x_{N_n}$ of points in D such that for any $x \in D$, there exists a grid point x_k with $|x - x_k| \leq \delta_n$. How many points N_n are required if we choose a spacing of δ_n ?

8. *Lipschitz Continuity of the Process.* Using the Lipschitz property of the kernel K , find a bound for the difference $|Z_n(x) - Z_n(x')|$ in terms of $|x - x'|$. Specifically, show that:

$$|Z_n(x) - Z_n(x')| \leq \frac{C_L}{h_n^2} |x - x'|$$

for some constant C_L independent of x, n, h_n .

9. *The Union Bound.* Show that

$$\sup_{x \in D} |Z_n(x)| \leq \max_{k=1, \dots, N_n} |Z_n(x_k)| + \sup_{x, x': |x-x'| < \delta_n} |Z_n(x) - Z_n(x')|,$$

and choose δ_n (as a function of h_n, n, ϵ) such that the second term (the discretization error) is strictly less than $\epsilon/2$. Apply the Union Bound and the Bernstein result (Part 6) to the first term (the grid maximum).

10. *Conclusion.* Determine the sufficient condition on the rate of decay of h_n for uniform consistency, that is, $\sup_{x \in D} |\hat{f}_n(x) - f(x)| = o_{\mathbb{P}}(1)$.

Table 1.1. Discrete Distributions on $\mathcal{R}$

Uniform	p.d.f.	$1/m, x = a_1, \dots, a_m$
	m.g.f.	$\sum_{j=1}^m e^{a_j t}/m, t \in \mathcal{R}$
$DU(a_1, \dots, a_m)$	Expectation	$\sum_{j=1}^m a_j/m$
	Variance	$\sum_{j=1}^m (a_j - \bar{a})^2/m, \bar{a} = \sum_{j=1}^m a_j/m$
	Parameter	$a_i \in \mathcal{R}, m = 1, 2, \dots$
Binomial	p.d.f.	$\binom{n}{x} p^x (1-p)^{n-x}, x = 0, 1, \dots, n$
	m.g.f.	$(pe^t + 1 - p)^n, t \in \mathcal{R}$
$Bi(p, n)$	Expectation	np
	Variance	$np(1-p)$
	Parameter	$p \in [0, 1], n = 1, 2, \dots$
Poisson	p.d.f.	$\theta^x e^{-\theta}/x!, x = 0, 1, 2, \dots$
	m.g.f.	$e^{\theta(e^t - 1)}, t \in \mathcal{R}$
$P(\theta)$	Expectation	θ
	Variance	θ
	Parameter	$\theta > 0$
Geometric	p.d.f.	$(1-p)^{x-1}p, x = 1, 2, \dots$
	m.g.f.	$pe^t/[1 - (1-p)e^t], t < -\log(1-p)$
$G(p)$	Expectation	$1/p$
	Variance	$(1-p)/p^2$
	Parameter	$p \in [0, 1]$
Hyper-geometric	p.d.f.	$\binom{n}{x} \binom{m}{r-x} / \binom{N}{r}$
		$x = 0, 1, \dots, \min\{r, n\}, r - x \leq m$
$HG(r, n, m)$	m.g.f.	No explicit form
	Expectation	rn/N
	Variance	$rn m(N-r)/[N^2(N-1)]$
	Parameter	$r, n, m = 1, 2, \dots, N = n + m$
Negative binomial	p.d.f.	$\binom{x-1}{r-1} p^r (1-p)^{x-r}, x = r, r+1, \dots$
	m.g.f.	$p^r e^{rt}/[1 - (1-p)e^t]^r, t < -\log(1-p)$
$NB(p, r)$	Expectation	r/p
	Variance	$r(1-p)/p^2$
	Parameter	$p \in [0, 1], r = 1, 2, \dots$
Log-distribution	p.d.f.	$-(\log p)^{-1} x^{-1} (1-p)^x, x = 1, 2, \dots$
	m.g.f.	$\log[1 - (1-p)e^t]/\log p, t \in \mathcal{R}$
$L(p)$	Expectation	$-(1-p)/(p \log p)$
	Variance	$-(1-p)[1 + (1-p)/\log p]/(p^2 \log p)$
	Parameter	$p \in (0, 1)$

All p.d.f.'s are w.r.t. counting measure.

Table 1.2. Distributions on $\mathcal{R}$ with Lebesgue p.d.f.'s

Uniform	p.d.f.	$(b-a)^{-1}I_{(a,b)}(x)$
	m.g.f.	$(e^{bt} - e^{at})/[(b-a)t], \quad t \in \mathcal{R}$
$U(a, b)$	Expectation	$(a+b)/2$
	Variance	$(b-a)^2/12$
	Parameter	$a, b \in \mathcal{R}, \quad a < b$
Normal	p.d.f.	$\frac{1}{\sqrt{2\pi}\sigma}e^{-(x-\mu)^2/2\sigma^2}$
	m.g.f.	$e^{\mu t + \sigma^2 t^2/2}, \quad t \in \mathcal{R}$
$N(\mu, \sigma^2)$	Expectation	μ
	Variance	σ^2
	Parameter	$\mu \in \mathcal{R}, \quad \sigma > 0$
Exponential	p.d.f.	$\theta^{-1}e^{-(x-a)/\theta}I_{(a,\infty)}(x)$
	m.g.f.	$e^{at}(1-\theta t)^{-1}, \quad t < \theta^{-1}$
$E(a, \theta)$	Expectation	$\theta + a$
	Variance	θ^2
	Parameter	$\theta > 0, \quad a \in \mathcal{R}$
Chi-square	p.d.f.	$\frac{1}{\Gamma(k/2)2^{k/2}}x^{k/2-1}e^{-x/2}I_{(0,\infty)}(x)$
	m.g.f.	$(1-2t)^{-k/2}, \quad t < 1/2$
χ_k^2	Expectation	k
	Variance	$2k$
	Parameter	$k = 1, 2, \dots$
Gamma	p.d.f.	$\frac{1}{\Gamma(\alpha)\gamma^\alpha}x^{\alpha-1}e^{-x/\gamma}I_{(0,\infty)}(x)$
	m.g.f.	$(1-\gamma t)^{-\alpha}, \quad t < \gamma^{-1}$
$\Gamma(\alpha, \gamma)$	Expectation	$\alpha\gamma$
	Variance	$\alpha\gamma^2$
	Parameter	$\gamma > 0, \quad \alpha > 0$
Beta	p.d.f.	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}x^{\alpha-1}(1-x)^{\beta-1}I_{(0,1)}(x)$
	m.g.f.	No explicit form
$B(\alpha, \beta)$	Expectation	$\alpha/(\alpha+\beta)$
	Variance	$\alpha\beta/[(\alpha+\beta+1)(\alpha+\beta)^2]$
	Parameter	$\alpha > 0, \quad \beta > 0$
Cauchy	p.d.f.	$\frac{1}{\pi\sigma}\left[1+\left(\frac{x-\mu}{\sigma}\right)^2\right]^{-1}$
	ch.f.	$e^{\sqrt{-1}\mu t - \sigma t }$
$C(\mu, \sigma)$	Expectation	Does not exist
	Variance	Does not exist
	Parameter	$\mu \in \mathcal{R}, \quad \sigma > 0$

Table 1.2. (continued)

t-distribution	p.d.f.	$\frac{\Gamma[(n+1)/2]}{\sqrt{n\pi}\Gamma(n/2)} \left(1 + \frac{x^2}{n}\right)^{-(n+1)/2}$
	ch.f.	No explicit form
t_n	Expectation	0, ($n > 1$)
	Variance	$n/(n-2)$, ($n > 2$)
	Parameter	$n = 1, 2, \dots$
F-distribution	p.d.f.	$\frac{n^{n/2}m^{m/2}\Gamma[(n+m)/2]x^{n/2-1}}{\Gamma(n/2)\Gamma(m/2)(m+nx)^{(n+m)/2}}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$F_{n,m}$	Expectation	$m/(m-2)$, ($m > 2$)
	Variance	$2m^2(n+m-2)/[n(m-2)^2(m-4)]$, ($m > 4$)
	Parameter	$n = 1, 2, \dots$, $m = 1, 2, \dots$
Log-normal	p.d.f.	$\frac{1}{\sqrt{2\pi}\sigma}x^{-1}e^{-(\log x - \mu)^2/2\sigma^2}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$LN(\mu, \sigma^2)$	Expectation	$e^{\mu+\sigma^2/2}$
	Variance	$e^{2\mu+\sigma^2}(e^{\sigma^2} - 1)$
	Parameter	$\mu \in \mathcal{R}$, $\sigma > 0$
Weibull	p.d.f.	$\frac{\alpha}{\theta}x^{\alpha-1}e^{-x^\alpha/\theta}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$W(\alpha, \theta)$	Expectation	$\theta^{1/\alpha}\Gamma(\alpha^{-1} + 1)$
	Variance	$\theta^{2/\alpha} \left\{ \Gamma(2\alpha^{-1} + 1) - [\Gamma(\alpha^{-1} + 1)]^2 \right\}$
	Parameter	$\theta > 0$, $\alpha > 0$
Double Exponential	p.d.f.	$\frac{1}{2\theta}e^{- x-\mu /\theta}$
	m.g.f.	$e^{\mu t}/(1 - \theta^2 t^2)$, $ t < \theta^{-1}$
$DE(\mu, \theta)$	Expectation	μ
	Variance	$2\theta^2$
	Parameter	$\mu \in \mathcal{R}$, $\theta > 0$
Pareto	p.d.f.	$\theta a^\theta x^{-(\theta+1)}I_{(a,\infty)}(x)$
	ch.f.	No explicit form
$Pa(a, \theta)$	Expectation	$\theta a/(\theta - 1)$, ($\theta > 1$)
	Variance	$\theta a^2/[(\theta - 1)^2(\theta - 2)]$, ($\theta > 2$)
	Parameter	$\theta > 0$, $a > 0$
Logistic	p.d.f.	$\sigma^{-1}e^{-(x-\mu)/\sigma}/[1 + e^{-(x-\mu)/\sigma}]^2$
	m.g.f.	$e^{\mu t}\Gamma(1 + \sigma t)\Gamma(1 - \sigma t)$, $ t < \sigma^{-1}$
$LG(\mu, \sigma)$	Expectation	μ
	Variance	$\sigma^2\pi^2/3$
	Parameter	$\mu \in \mathcal{R}$, $\sigma > 0$